Lubomír Bureš

Nuclear engineer specialising in computational modelling and technical leadership.

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Professional Experience

Saltfoss Energy (formerly Seaborg Technologies) *Copenhagen, Denmark*
Senior Multiphysics Specialist March 2024 – present

- Serve as a system technical lead within molten-salt reactor development, guiding design work for the system and coordinating with other system leads.
- Establish modelling approaches and develop tools for engineering analysis, then support colleagues in taking ownership while remaining available for technical review and guidance.
- Combine engineering leadership with hands-on modelling and analysis across reactor physics and thermal hydraulics, including development with the MOOSE Framework.
- Publish selected research and present technical developments at scientific conferences and industry forums.

Multiphysics Specialist October 2021 – March 2024

- Developed and applied Python and Modelica models for molten-salt reactor physics, thermal hydraulics, and system dynamics.
- Supervised a master's thesis project whose candidate subsequently joined the company as a full-time employee.
- Published peer-reviewed research and represented the company both in Denmark and internationally.

Paul Scherrer Institute *Villigen, Switzerland*
Doctoral Researcher October 2018 – September 2021

- Developed numerical methods and high-fidelity simulations of phase change and nucleate boiling.
- Implemented numerical methods in the C++ PSI-BOIL research code; verified and validated models and investigated microlayer dynamics.
- Supervised four student projects, resulting in published research.
- Led exercise sessions for a course at ETH Zurich.

Supplementary Projects

Winterlead *Remote*
Independent Consultant April 2024 – present

- Completed projects in software development and computational engineering, including software for Quantified Carbon and volunteer technical contributions to the Repower Initiative.

Education

EPF Lausanne, Institute of Physics *Paul Scherrer Institute (PSI), Villigen, Switzerland*
Doctor of Science (PhD) 2018 – 2021

- *Thesis*: Fundamental study on microlayer dynamics in nucleate boiling
- *Achievements*: Awarded the 2022 PhD Prize by the European Nuclear Education Network Association.

ETH Zurich – EPF Lausanne *Zürich & Lausanne, Switzerland*
Joint MSc in Nuclear Engineering 2016 – 2018

- *Achievements*: Graduated with distinction. Awarded the EPFL Excellence Fellowship over the duration of the study programme.
- *Certifications*: Received the European Master of Science in Nuclear Engineering certification.

Czech Technical University *Prague, Czech Republic*
BSc in Nuclear Engineering 2013 – 2016

- *Achievements*: Graduated with honours.

Skills

Nuclear engineering:	Reactor physics, thermal hydraulics, molten-salt reactor systems
Computational modelling:	Multiphysics simulation, numerical methods, model verification and validation
Languages and frameworks:	Python, Modelica, MOOSE Framework

Languages

Czech (native), English (C2), German (C1), Danish (B2)

Organisational Involvement

IDA Nuclear *Danish Society of Engineers, Denmark*
Board Member September 2025 – present

- Organise events promoting knowledge and education about nuclear and atomic technology.

Kildevæld Kulturcenter, Digitalt Værksted *Copenhagen, Denmark*
Volunteer / Superuser October 2025 – present

- Support open workshop sessions by helping citizens use digital fabrication tools, including 3D printing and laser cutting, and by discussing and developing project ideas.

PhD and Postdoc Association *Paul Scherrer Institute, Switzerland*
President October 2019 – May 2021

- Led the association through the Covid-19 pandemic, oversaw bylaws enabling non-profit registration, and represented members within and outside PSI.
- March 2019 to September 2019: Social Committee Head.

Various *Europe-wide*
Volunteer / Event Organiser October 2018 – November 2023

- Volunteered for and helped organise multiple climate-change-related events across Europe.
- Represented the Nuclear for Climate initiative at the UN Climate Change Conference COP25 in Madrid.